

Computer Science and Engineering Department
Motilal Nehru National Institute of Technology Allahabad
B.Tech IV Semester
Operating System Lab (CSN14401)
Assignment 9

1. Assume that a system has a 32-bit virtual address with a 4-KB page size. Write a C program that is passed a virtual address (in decimal) on the command line and have it output the page number and offset for the given address. As an example, your program would run as follows:
./a.out 19986
Your program would output:
The address 19986 contains:
page number = 4
offset = 3602
Writing this program will require using the appropriate data type to store 32 bits. We encourage you to use unsigned data types as well.
2. Write a program that implements the FIFO and LRU page-replacement algorithms. First, generate a random page reference string where page numbers range from 0 to 9. Apply the random page reference string to each algorithm, and record the number of page faults incurred by each algorithm. Implement the replacement algorithms so that the number of page frames can vary from 1 to 7. Assume that demand paging is used.